Enhanced PMOS NBTI Degradation Due to Halo Implant Channeling

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ABSTRACT

NBTI is a serious reliability concern in state of the art PMOSFET devices. The implementation of nitrided gate oxides to prevent boron penetration has aggravated the NBTI issue. Because of relaxation effects careful stress and measurement techniques ("On-the-Fly") must be used for reliable estimation of device lifetime. This abstract describes a unique enhanced NBTI degradation phenomenon in which NBTI induced V_T degradation was determine to be directly proportional to the initial V_T of the device. Electrical results from special test structures identified halo implant channeling as causing the enhanced NBTI induced degradation behavior.
[Kewords: NBTI, halo implant]

I. INTRODUCTION

NBTI is a serious reliability concern in state-of-the-art PMOSFET devices [1-11]. The implementation of nitrided gate oxide to prevent boron penetration aggravates the NBTI issue [1,2,5]. Because of recovery effects [14-16] careful stress and measurement techniques ("On-the-Fly") must be used for reliable estimation of device lifetime [3, 12-14]. The present theory of NBTI for PMOS devices states that under negative gate bias and elevated temperature, inversion layer holes interact with hydrogen passivated silicon bonds to break the Si-H bonds creating donor like interface traps and H_2 atoms that diffuse from the surface. On removal of the stress the H_2 atoms can diffuse back to the surface repassivating an existing trap (recovery effect). NBTI induced interface trap generation has been modeled using Reaction-Diffusion theory [17] which predicts NBTI V_T degradation to display a power law dependence on stress time of 1/6 [18].

This work focuses on a DGO PMOS device that exhibited enhanced NBTI degradation behavior. Through the use of special test structures and wafer level multi-site NBTI stress methods it is determined that halo implant channeling is responsible for the enhanced NBTI degradation behavior. A model is proposed, whereby, halo implant channeling through the polysilicon gate neutralizes the PLDD implant extending the device channel under the device spacer. This increases the number of Si-H bonds enhancing the NBTI degradation behavior of the device.

II. EXPERIMENTAL DETAILS

Fig.1 displays the cross-section of the 3V W/L = 10/0.3 µm DGO PMOS device used in this study. This device has self-aligned angled halo implants at the source and drain ends of the device to prevent device punchthrough. In addition, to prevent halo implant channeling a thin oxide layer (Halo Implant Oxide) is grown over the poly-silicon gate just prior to the halo implant.

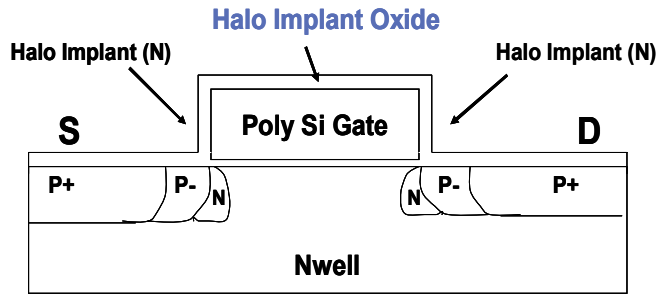

Fig. 1. DGO 3V PMOS device showing self-aligned N-type angled halo implant and halo implant oxide to prevent implant channeling.

Several test methods will be used to characterize the enhanced NBTI degradation phenomena. The "On-the-Fly" continuous stress method [3] will be used to characterize the 500 hr NBTI degradation. In this case a small V_{DS} (-0.05V) is applied to the drain during stress (V_{GS} stress = -3.3V, T = 150°C) and the drain current (I_D) monitored continuously. The V_T degradation is determined using the following expression $\Delta V_T = \Delta I_D/I_{D0} * (V_{GS} - V_{T0})$, where I_{D0} is the initial measured drain current, ΔI_D the change in the measured drain current during stress (I_D - I_{D0}), V_{TO} the initial threshold voltage of the device and V_{GS} the applied stress gate voltage (-3.3V). In this method it is critical to accurately determine I_{D0} in the shortest time possible. We use fixed measurement range and zero time measurement delays to obtain the initial I_{DO} within 5 ms of applying the stress. In addition, gate stressing is done at a conservative -3.3V just 10% above the device use bias condition.

The second experimental NBTI method implements a fast V_T search I_D (I_D target = -0.1µA/µm, V_{DS} = -0.1V) algorithm with zero measurement time delay to determine the V_T of 6 identical PMOS devices within a test quad. V_T measurements are made before and after a fixed V_{GS} = -3.6V, T = 130°C, 1ks stress. This NBTI stress algorithm is implemented in an automated prober environment so that multiple wafer site and wafer NBTI measurements can easily be made. Finally, variable base (V_{BASE} = -3V to 1V, 40 points) fixed amplitude (2V) charge pumping current (Icp) measurements were made to characterize the fast interface state density (Nit) before and after a fixed V_{GS} = -3.9V, T = 35°C, 10 ks NBTI stress. Though partial V_T recovery is expected when utilizing the last two measurement techniques

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(fast V_T search and charge pumping), these test results are used to compare device and wafer split results and not as an absolute measure of NBTI degradation.

III. Experimental Results

Fig. 2 displays typical "On-the-Fly" degradation results vs. stress time from the device displayed in Fig. 1 stressed at a V_{GS} = -3.3V and temperature of 150°C. These results show that the degradation is significant ~160 mV in 500 hrs (expected degradation ~75 mV). It was also observed that the initial $|V_T|$ of the device 0.68V was anomalously high when compared to the expected target value 0.5V.

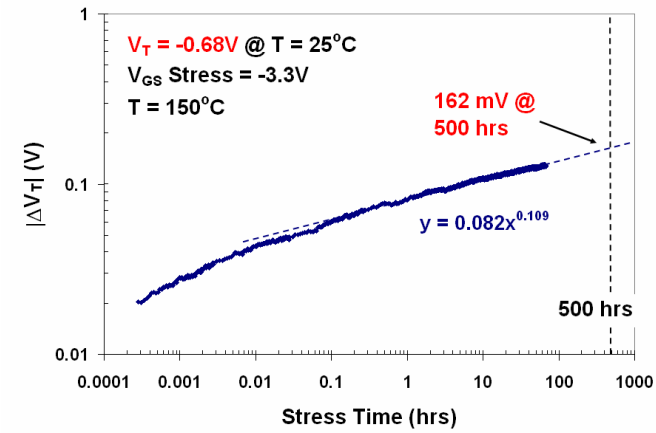
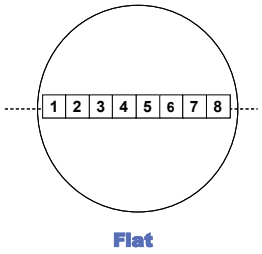


Fig. 2. "On-the-Fly" V_T degradation (ΔV_T) results displaying enhanced NBTI degradation.

In order to better understand the enhanced NBTI degradation and high $|V_T|$ results eight wafer test sites were chosen as shown in Fig. 3 for further V_T measurements. Each test site contains 6 identical devices within a single test quad (in total 48 devices were tested). Fig. 3 shows tabulated initial linear V_T @ V_{DS} = -0.1 for the 6 transistors (V_T1-V_T6) vs. test site position (test sites 1-8). These results show not only large variation in the initial V_T ~ 300 mV across the wafer but large variation is seen even in adjoining devices on a single test quad where distances between devices are ~300 µm. To study the effect of this high V_T variability on NBTI degradation the lowest V_T device (first device on test site 3: V_T = -0.41 V) was selected for further NBTI testing.



(a)

Site	V_T1	V_T2	V_T3	V_T4	V_T5	V_T6
1	-0.44	-0.59	-0.60	-0.66	-0.44	-0.44
2	-0.56	-0.52	-0.43	-0.41	-0.58	-0.57
3	-0.41	-0.63	-0.56	-0.45	-0.48	-0.58
4	-0.56	-0.62	-0.60	-0.53	-0.62	-0.62
5	-0.64	-0.46	-0.48	-0.45	-0.56	-0.61
6	-0.57	-0.63	-0.56	-0.46	-0.64	-0.49
7	-0.62	-0.67	-0.67	-0.65	-0.52	-0.58
8	-0.67	-0.68	-0.68	-0.66	-0.56	-0.64

(b)

Fig. 3. Selected sites (a) and measured initial threshold voltage (V_T) @ 25°C for 6 identical devices (V_T1-V_T6) as a function of test site (b). Large variation is seen in initial V_T (~0.2 V) even for adjoining devices.

Fig. 4 displays "On-the-Fly" ΔV_T degradation results from the V_T = -0.41 V device. These results show that the degradation is significantly reduced (88 mV vs. 162 mV) than the device (V_T = -0.68V) results shown in Fig. 2. Since the initial device V_T and the ΔV_T degradation seem to be correlated an automated NBTI test program was generated that measures the V_T degradation on multiple wafer sites after 1 ks NBTI stress. These results are seen in Fig. 5 and show that the initial device V_T and the NBTI induced V_T degradation are strongly correlated (R^2 = 0.90). Resolving the V_T variability should, therefore, resolve the enhanced NBTI issue.

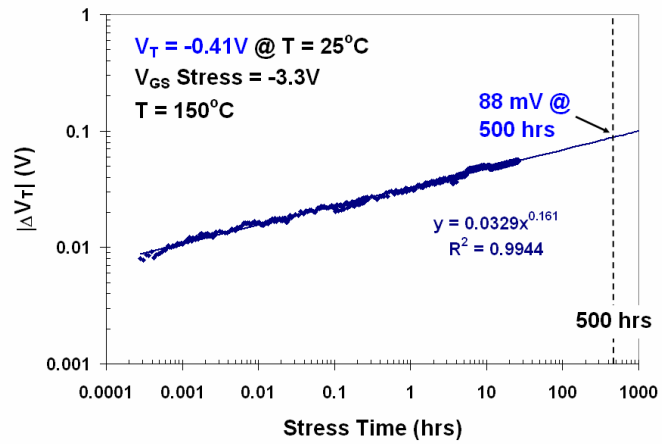


Fig. 4. "On-the-Fly" V_T degradation (ΔV_T) results from Low V_T device (V_T = -0.41 V).

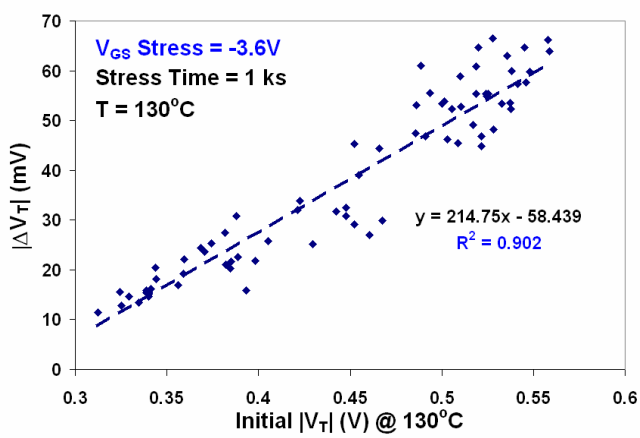


Fig. 5. V_T degradation after 1 ks stress vs. measured initial V_T @ 130°C showing that the NBTI degradation and initial V_T are correlated.

To study the origin of the large V_T variation and ΔV_T, two special test structures were created as seen in Fig. 6. The top device has a drain side only halo (DOH) implant while the bottom device (no halo implant) was processed with a blanket anti-punchthrough (N^-_{APT}) implant. A significant difference between the devices shown in Fig. 6 is that the halo implant is performed after gate oxide growth while the blanket N^-_{APT} implant is performed before gate oxide growth. Fig. 7 displays NBTI induced ΔV_T degradation results vs. initial V_T from the two devices shown in Fig. 6. These results show that the DOH device displays enhanced NBTI degradation while

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the no halo device shows minimal ΔV_T degradation, V_T variation and no correlation of ΔV_T with initial V_T.

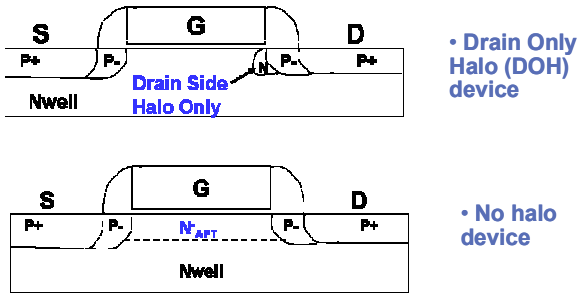


Fig. 6. To understand the anomalous NBTI and V_T behavior the Drain Only Halo (DOH) and no halo devices were created.

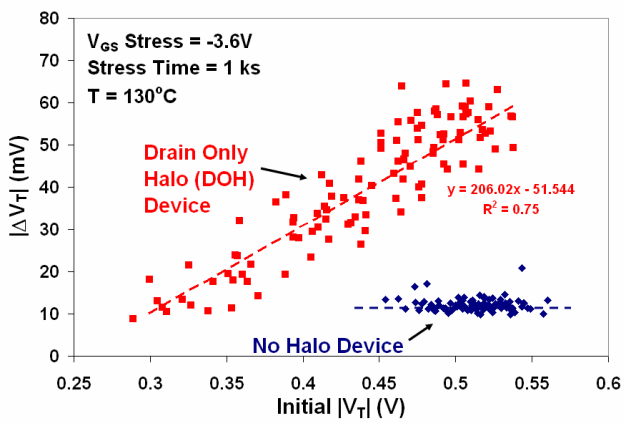


Fig. 7. NBTI results from DOH and no halo device show that only the DOH device displays high V_T variability and anomalous NBTI degradation.

In order to understand the effect of the halo implant on V_T variability drain current (I_D) measurements were made on DOH device as a function of drain bias. Device simulations in Fig. 8 show that at low drain bias (V_{DS} = -0.1 V) the hole current is along the surface. In this case the entire channel determines the V_T of the device while in the case of high $|V_{DS}|$ the hole current deviates from the surface when it enters the depletion region and is, therefore, not affected by interface states (Nit) or fixed oxide traps (Not) within this region. In the case of non-uniformly doped channels the device V_T is determined by the part of the channel that inverts at the highest magnitude value of V_T. I_D vs. V_{DS} measurements can, therefore, be used to determine V_T (charge profiles) vs. channel position.

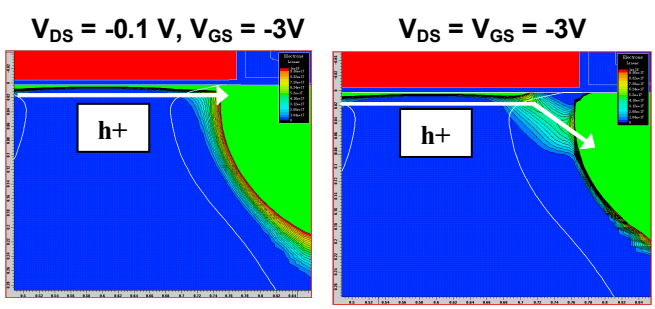
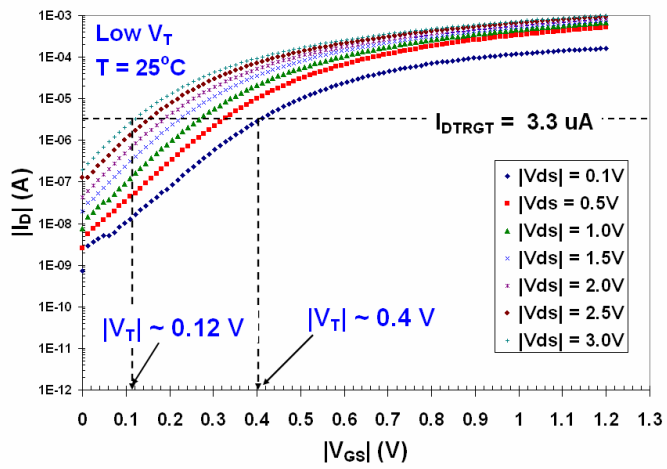
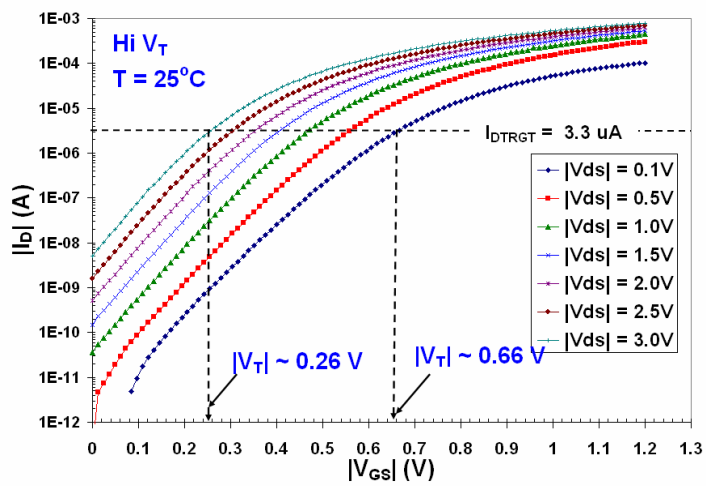


Fig. 8. Simulated hole current vs. V_{DS} demonstrating V_T profile measurement technique.

Fig. 9 displays I_D vs. V_{DS} for the DOH Low and Hi $|V_T|$ devices. These results show that both the Low and Hi V_T device I_D-V_G curves shift significantly to left with increased $|V_{DS}|$ consistent with the highest channel $|V_T|$ being at the drain (halo implant) end of the device.



(a)



(b)

Fig. 9. Low (a) and Hi V_T (b) drain current (I_D) measurements vs. V_{GS}. I_D curves for both devices shift significantly to the left with increased $|V_{DS}|$ consistent with the halo implant at the drain end of the device determining the device V_T.

Fig. 10 displays forward and reverse (in the reverse mode the drain is grounded and a source voltage (V_{SD}) is applied to the device) V_T @ 3.3μA vs. $(V_{DS})^{0.5}$ for the Low and Hi $|V_T|$ DOH devices. These results show that in the forward case (drain side V_{DS}) the High V_T device's V_T roll off is much faster (larger slope) than the Low V_T device. These results suggest that halo implant channeling is responsible for the large V_T variability and potentially the enhanced NBTI degradation behavior.

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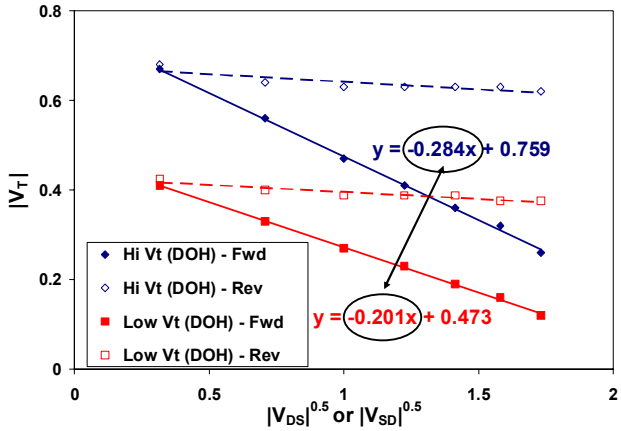
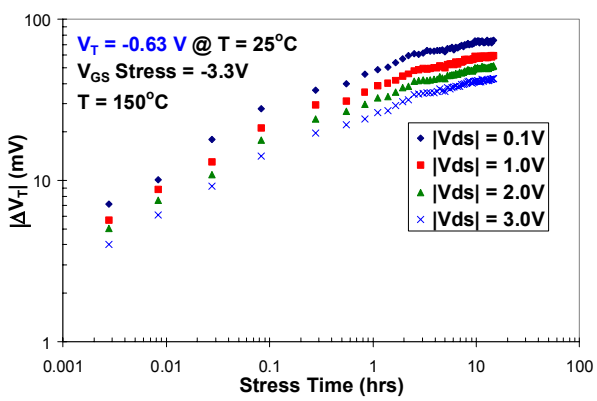
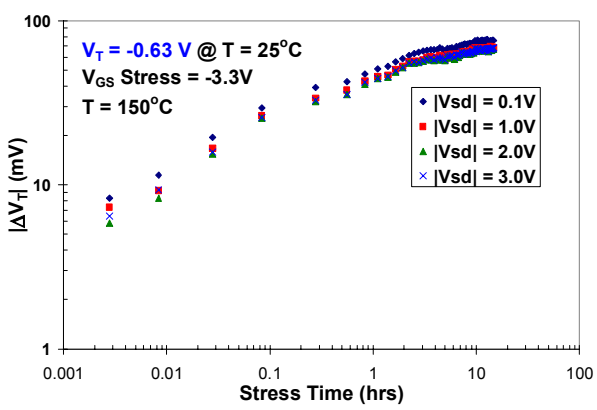


Fig. 10. Forward and reverse V_T profile results show that the Hi V_T device's V_T decreases more rapidly indicating that the drain edge halo implant is responsible for the observed large V_T variations.

Fig. 11 displays forward and reverse NBTI degradation results for the DOH device. In these measurements the forward $|V_{DS}|$ or the source $|V_{SD}|$ was stepped from 0.1V to 3.0 V. These results show that in the forward measurement mode the NBTI induced $|\Delta V_T|$ decreases with $|V_{DS}|$ while for the reverse case the $|\Delta V_T|$ is nearly independent of $|V_{SD}|$. These results again strongly suggest that the halo implant is responsible for the enhanced NBTI degradation behavior.



(a)



(b)

Fig. 11. Forward (a) and reverse (b) NBTI induced ΔV_T degradation results vs. $|V_{DS}|$ or $|V_{SD}|$.

To prevent halo implant channeling the halo implant oxide thickness just prior to halo implantation was increased (see Fig. 12). Increasing oxide thickness results in increased ion scattering reducing the likelihood of polysilicon gate channeling. It was determined empirically that increasing the halo implant oxide thickness by ~8X greatly reduced the V_T variability.

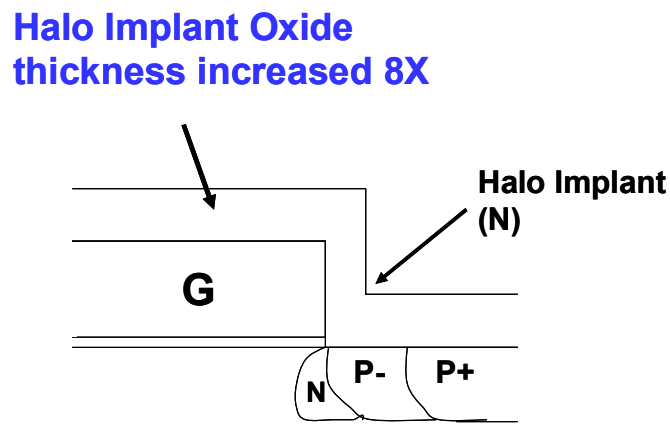


Fig. 12. It was determined that increasing the halo implant oxide thickness prior to halo implant by ~8X resolved the V_T variability.

Fig. 13 compares fast NBTI results from devices fabricated with the original and 8X halo implant oxide thickness. These results show that devices fabricated with the thicker halo oxide display reduced V_T variability and NBTI induced ΔV_T degradation. Fig. 14 displays typical "On-the-Fly" V_T degradation vs. stress time results from an 8X halo implant oxide device. These results show that the V_T degradation is reduced to 75 mV vs. the original High V_T device 162 mV @ 500 hrs (see Fig. 2).

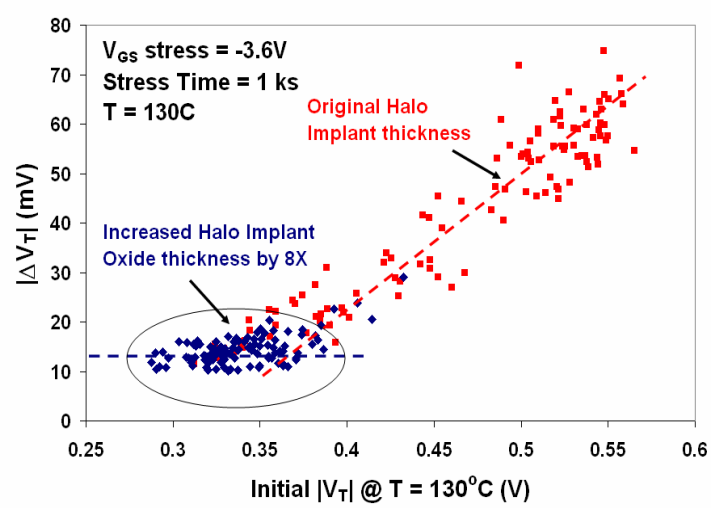


Fig. 13. Effect of increased halo implant oxide thickness on NBTI induced ΔV_T degradation. ΔV_T and V_T variation is significantly reduced.

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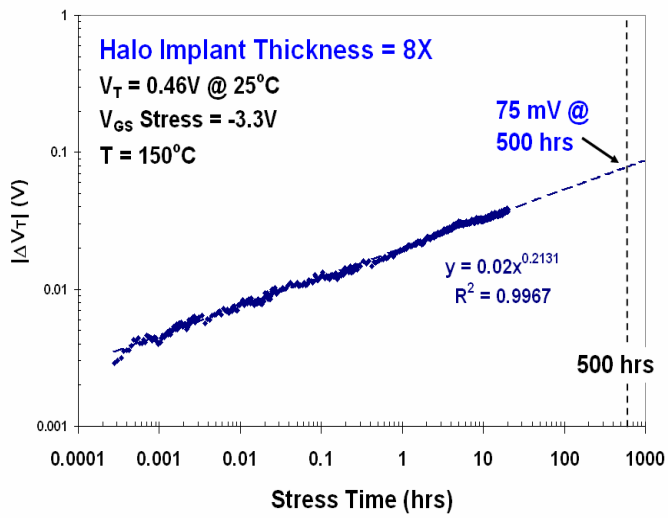


Fig 14. "On-the-Fly" ΔV_T results from a device with the 8X halo implant thickness displaying reduced V_T degradation compared to High V_T device (compare to Fig. 2).

Fig. 15 displays forward maximum (peak) substrate current (Isubmax) results vs. initial V_T for a DOH device. These results compare maximum substrate results from devices with the original and 8X halo implant oxide thickness. Fig. 15 shows that Isubmax increases with increased initial V_T (opposite to what would be expected). These results are consistent with a degradation model where channeling halo N-type implant neutralizes the PLDD implant causing the drain junction to be more abrupt (see Fig. 1) moving the device channel under the spacer. For the 8X halo implant device V_T and Isubmax correlation still exists but the Isubmax is greatly reduced indicating that less PLDD neutralization has occurred.

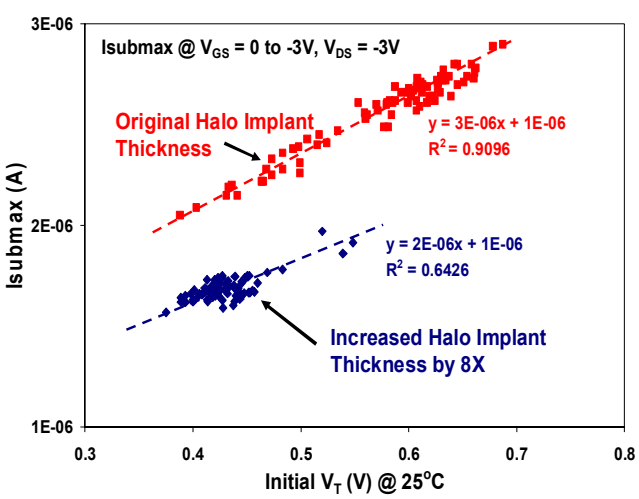
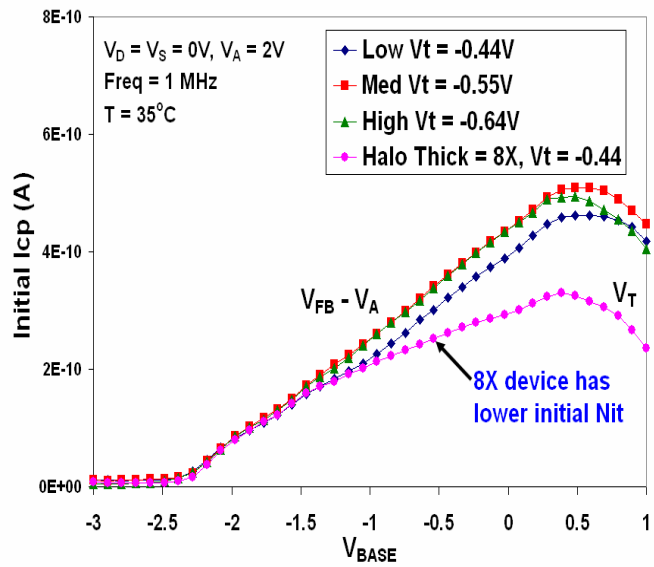
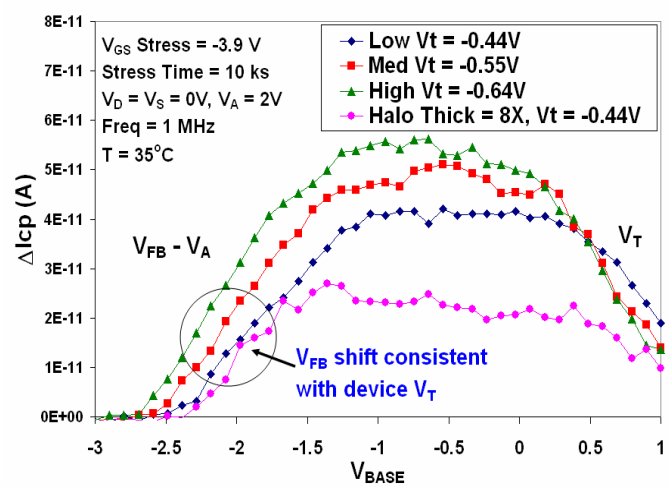


Fig. 15. Effect of 8X halo implant oxide thickness on drain side maximum substrate current (Isubmax).

Fig. 16 displays typical initial (a) charge pumping current Icp and degradation (ΔIcp) (b) results after a 10 ks NBTI stress comparing the original halo implant oxide Low, Med and High V_T devices with the 8X halo implant oxide device. These results show that devices fabricated with the original halo implant oxide thickness have greater initial Icp (fast interface traps - Nit) and ΔIcp when compared to the device fabricated with the 8X halo implant oxide.



(a)



(b)

Fig 16. Typical initial Icp (a) and degradation (ΔIcp) results (b) from the original Low, Med, High V_T devices and the 8X halo oxide device.

A potential model explaining the phenomena is displayed in Fig. 17. Halo implant channeling through the polysilicon gate neutralizes the PLDD implant extending the device channel under the spacer exposing a large number of additional Si-H bonds created during spacer processing. The device V_T is increased because the extended device channel effectively increases the gate oxide thickness and halo implant channeling increases the N-type dopant at the Si-SiO_2 surface near the device edge. The NBTI degradation is enhanced because of the additional exposed Si-H bonds under the device spacer.

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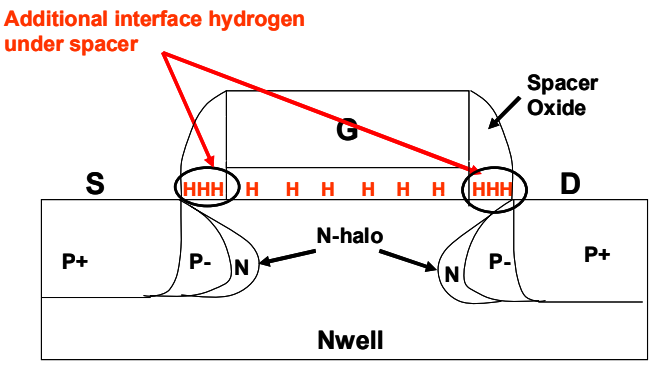

Fig. 17. N-type halo implant channeling causes V_T variability due to neutralization of the PLDD implant extending the device channel under the spacer effectively increasing the oxide thickness and the N-type dopant at the Si-SiO_2 interface. The NBTI degradation is enhanced because of the large number of additional exposed Si-H bonds under the device spacer.

The model shown in Fig. 17 explains the anomalous V_T and enhanced NBTI degradation in terms of halo implant channeling neutralizing the PLDD implant at the source and drain end of the devices. Fig. 18 displays V_T degradation results from the original device architecture with minimum halo oxide implant thickness where the PLDD dose is increased up to 4X the original dose (note: the halo dose is 80% the normal dose). These results show that the anomalous V_T variability and ΔV_T is greatly reduced for increased PLDD dose consistent with the model described in Fig. 17.

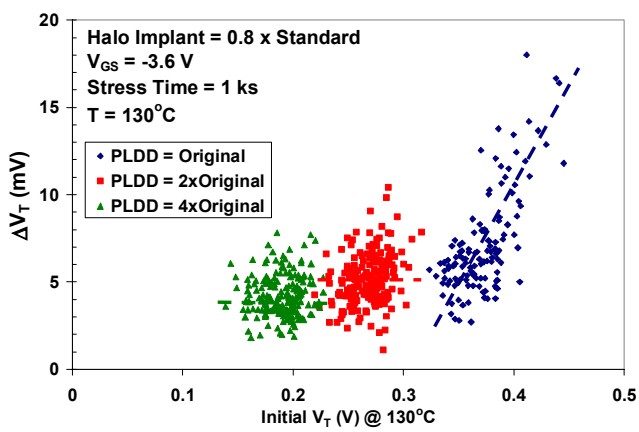

Fig. 18. Increasing the PLDD dose reduces the V_T variability and ΔV_T consistent with the model described in Fig. 17.

IV. Summary and Conclusions

Anomalously high NBTI induced degradation and V_T variation was observed for the 3V PMOS DGO device that correlated to initial device V_T. V_T profile and NBTI degradation results from special test structures (drain only halo and no halo) identified the halo implant and halo implant channeling as the potential source of the anomalous NBTI behavior. To reduce channeling the halo implant oxide thickness just prior to halo implantation was increased by 8X. Devices fabricated with this process change displayed significantly reduced NBTI induced V_T degradation, charge pumping current and V_T variability. An NBTI degradation model is proposed which is based on halo implant channeling neutralizing the PLDD implant extending the device channel under the spacer effectively increasing the oxide thickness, N-type Si-SiO_2 interface dopants and Si-H bonds. The NBTI degradation is enhanced because of the large number of additional exposed Si-H bonds under the device spacer.

References

[1] A. Sakuma et al., "Investigation of nitrogen originated NBTI mechanism in SiON with high nitrogen concentration", IRPS 2006, p. 454

[2] S.S. Tan et al., "Neighboring effect in nitrogen-enhanced negative bias temperature instability", SSDM, p.70, 2003.

[3] A. T. Krishnan et al., "Material dependence of hydrogen diffusion: implications for NBTI degradation", IEDM, p.688, 2005.

[4] D. Varghese et al., "On the dispersive versus arrhenius temperature activation of NBTI time evolution in plasma nitrided gate oxides: measurements, theory, and implications", IEDM, p.684, 2005.

[5] Y. Mitani, "Influence of nitrogen in ultra-thin SiON on negative bias temperature instability under AC stress", IEDM, p.117, 2004.

[6] A.T. Krishnan, "NBTI: Process, Device and Circuit", IRPS, , p.212, 2005 Tutorial.

[7] C. Schlunder,, "Mixed Signal Circuit Reliability – From Device Stress Conditions to IC-failure", IRPS, p.121, 2005 Tutorial.

[8] G La Rosa et al., "NBTI-channel hot carrier effects in PMOSFETs in advanced CMOS technologies", IRPS, p.282, 1997.

[9] D. Schroder et al., "Negative Bias Temperature Instability: Road to cross in deep submicron silicon semiconductor manufacturing", J. Appl. Phys. 94, p.1, 2002.

[10] V.Reddy et al., "Impact of Negative–Bias temperature instability on digital circuit reliability", IRPS, p.248, 2004.

[11] A.T. Krishnan et al., "NBTI Impact on Transistor and Circuit: Models , Mechanisms and Scaling Effects", IEDM, p.349, 2003.

[12] M. Denais et al., ""On-the-Fly characterization of NBTI in ultra-thin gate oxide PMOSFETS", IEDM 2004, P. 109

[13] H. Reisinger et al., "Analysis of NBTI degradation and recovery behavior based on ultra fast V_T measurements", IRPS 2006, p. 448

[14] C. R. Parthasarathy et al., "New insights into recovery characteristics post NBTI stress", IRPS, p.471, 2006.

[15] S. Rangan et al., "Universal recovery behavior of negative bias temperature instability", IEDM, p.341, 2003.

[16] T. Grasser et al., "The Universality of NBTI Relaxation and its Implication for Modeling and Characterization", IRPS, p.268, 2007.

[17] S. Chakravarthi et al., "A Comprehensive Framework for Predictive Modeling of Negative Bias Temperature Instability," *IRPS* 2004, pp. 273-282

[18] K. Kang et al., "Estimation of NBTI Degradation using I_{DDQ} Measurement," *IRPS* 2007, pp. 10-16

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